

# MOF H<sub>2</sub> Screening — Project Plan

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## Overview

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After looking at this more carefully, the right move is to keep scope tight and get a clean, focused paper out quickly rather than expand. The core asset is the work that's already done; what's needed is targeted polish plus two ML additions that strengthen the methods, then I run an updated GCMC validation pass on the workstation. Total target timeline is roughly two months end-to-end.

Phase 1 is your part: about four to five weeks, all from your laptop using your existing trained models, descriptor table, and code. Two milestones, two weeks each.

## Phase 1 — Your milestones

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### Milestone 1.1 — Interpretability

Weeks 1–2

#### Tasks

- Compute SHAP values on your trained model
- Compute permutation importance
- Generate a feature-contribution figure for the top 5 and bottom 5 ranked MOFs
- Begin drafting the Methods section of the manuscript in parallel

**Deliverable.** One notebook, two manuscript figures, a draft Discussion paragraph on feature importance, and the Methods section started.

### Milestone 1.2 — Uncertainty quantification + cross-database check

Weeks 3–4

#### Tasks

- Wrap the trained model with conformal prediction using the `MAPIE` library
- Generate 90% and 95% prediction intervals across all 12,986 MOFs
- Produce a coverage validation plot
- Apply the trained model to a QMOF subset for a generalization sanity check (no new GCMC; uses their reported uptakes as proxy)
- Continue the manuscript draft — aim for v1 of Methods, Results scaffolding, and Introduction by end of Week 4

**Deliverable.** Updated 12,986-MOF ranking with calibrated intervals, coverage figure, cross-database figure, and manuscript draft v1.

## End-of-Phase-1 handoff

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When Phase 1 finishes, please send across:

- Code repository as a tagged GitHub release with `environment.yml`

- Descriptor table for all 12,986 MOFs (CSV)
- Trained models and CV split indices (`.joblib` plus a JSON manifest)
- RASPA input templates and the force-field parameter files you used
- Manuscript draft v1 (Overleaf project link or Word document)

## After Phase 1

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I'll handle the GCMC work on the Ateneo workstation: re-activate the three low-performer reference structures using all-solvent-removed CIFs, and run an expanded validation pass on 20–30 additional MOFs sampled across the prediction range. Wall time is about two to three weeks. I'll send results in two batches so the manuscript keeps moving.

We finalize the manuscript together over a final two-week pass — you lead the Discussion, I draft the additional Methods and Results paragraphs from the new GCMC outputs and handle the cover letter and submission as corresponding author.

If anything in the milestone sequencing or effort estimates looks off when you start, push back. The plan is a draft, not a contract. There's no rush in the next week or two — take a real break first.